

MIS 637-C: DATA ANALYTICS AND MACHINE LEARNING

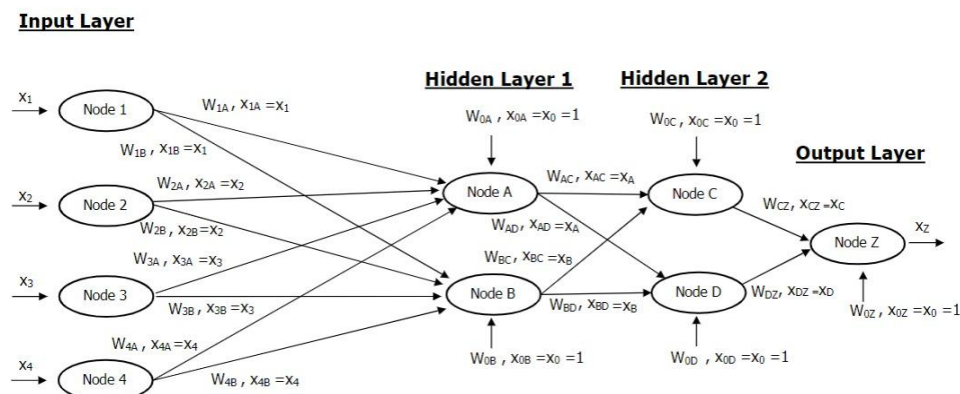
FINAL PROJECT REPORT

INTRODUCTION

A neural network consists of a layered, feedforward, completely connected network of artificial neurons, or nodes. The feedforward nature of the network restricts the network to a single direction of flow and does not allow looping or cycling. The neural network is composed of two or more layers, although most networks consist of three layers: an input layer, a hidden layer, and an output layer. There may be more than one hidden layer, although most networks contain only one, which is sufficient for most purposes. The neural network is completely connected, meaning that every node in a given layer is connected to every node in the next layer, although not to other nodes in the same layer. The neural network is completely connected, meaning that every node in a given layer is connected to every node in the next layer, although not to other nodes in the same layer. Each connection between nodes has a weight (e.g., W_{1A}) associated with it. At initialization, these weights are randomly assigned to values between zero and 1.

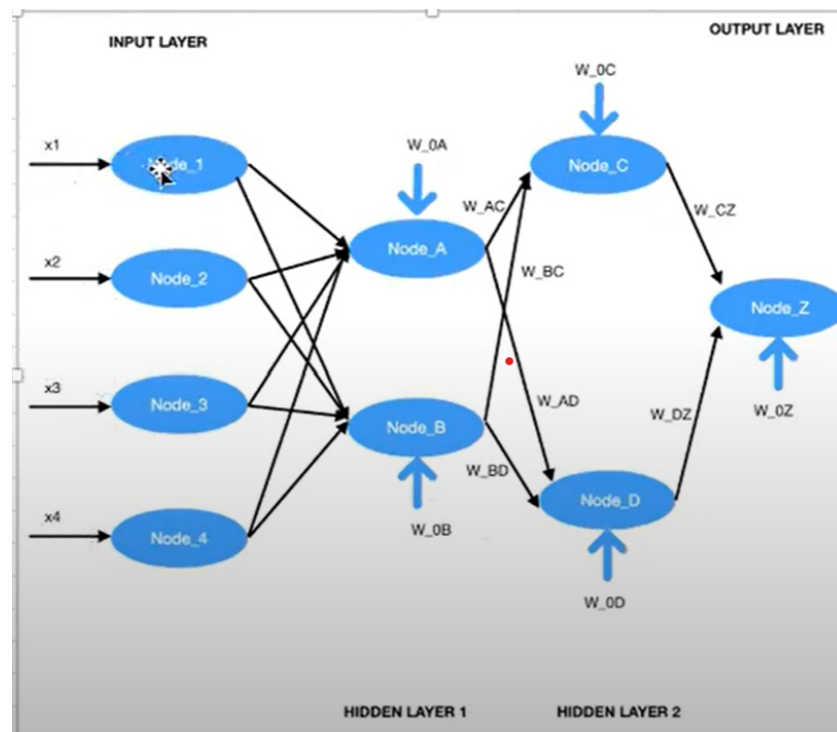
PROBLEM STATEMENT :

- There are 50 records presents in the given data. So, we divided the records equally among our group of 4 members (i.e; 12 to 13 records each)
- After dividing the records, the major problem we faced is that unless the first person completes the calculation of the set of records assigned to them, the second person cannot proceed with his part.
- This is a time consuming process and is also lengthy.



METHODOLOGIES :

- We are given a training set from x0 to x4 and also initial weights .We are using Excel sheet to make the calculations.Let us consider the calculation of record 5 from the excel sheet.So, to calculate the record 5 we need data from record 4.
- X1 to X4 are the inputs to the four nodes Node 1 , Node 2, Node 3 and Node 4 as shown below.



- Now we have to calculate the values of Net_A, Net_B, Net_C, Net_D, Net_Z. Lets discuss each calculation in detail.

- Calculation of Net_A :

– To calculate Net_A we use the below formulae ,

$$\text{Net_A} = X1 * W_{1A} + X2 * W_{2A} + X3 * W_{3A} + X4 * W_{4A} + X0 * W_{0A}$$

- After getting Net_A value, activate the node .To activate the function we should use the activation function also known as sigmoid function which is depicted as X_A .
- The sigmoid function formula to calculate X_A is ,

$$X_A = 1 / (1 + \exp(\text{Net}_A * -1))$$

- Similarly, we also calculate the value for Net_B

$$\text{Net}_B = X_1 * W_{1B} + X_2 * W_{2B} + X_3 * W_{3B} + X_4 * W_{4B} + X_0 * W_{0B}$$

- We also use the sigmoid function to calculate X_B the same as above to activate the nodes.

$$X_B = 1 / (1 + \exp(\text{Net}_B * -1))$$

- To calculate Net_C and Net_D we should use X_A and X_B .

$$\text{Net}_C = X_B * W_{BC} + X_A * W_{AC} + X_0 * W_{0C}$$

$$\text{Net}_D = X_B * W_{BD} + X_A * W_{AD} + X_0 * W_{0D}$$

- Now we should activate Net_C and Net_D using the sigmoid function.

$$X_C = 1 / (1 + \exp(\text{Net}_C * -1))$$

$$X_D = 1 / (1 + \exp(\text{Net}_D * -1))$$

- Now, X_C and X_D will be the inputs to Node Z .We calculate the Net_Z value and activate it which is depicted by X_Z .

$$X_Z = 1 / (1 + \exp(\text{Net_Z} * -1))$$

After we activate the final value of Net_Z we get the output of the entire neural network of Record 5.

- Now we need to calculate the delta value for each nodes. To calculate the delta value we should compare the X_Z value to the target variable value. We should calculate from forward to backward to get the delta values. To calculate delta_Z the formula is as follows,

$$\text{delta}_Z = X_Z * (1 - X_Z) * (\text{TV} - X_Z)$$

Here ,

TV = Target Variable

- Similarly, we calculate delta_A , delta_B , delta_C and delta_D .

$$\text{delta}_D = X_D * (1 - X_D) * (W_{DZ} * \text{delta}_Z)$$

$$\text{delta}_C = X_C * (1 - X_C) * (W_{CZ} * \text{delta}_Z)$$

$$\text{delta}_B = X_B * (1 - X_B) * (W_{DZ} * \text{delta}_Z) + (W_{BD} * \text{delta}_D)$$

$$\text{delta}_A = X_A * (1 - X_A) * (W_{CZ} * \text{delta}_Z) + (W_{AC} * \text{delta}_C)$$

- After calculating the delta values we will be able to calculate the resulting weights.
- To calculate the resulting weights we need to use the following formula,

$$W_{DZ} = \Delta_Z * \eta * X_D + \text{previous } W_{DZ}$$

$$W_{0Z} = \Delta_Z * \eta * X_0 + \text{previous } W_{0Z}$$

$$W_{0D} = \Delta_D * \eta * X_0 + \text{previous } W_{0D}$$

Here,

η = Learning Rate

- Similarly, we calculate the other weights for record 5. After calculating record 5 weights we will calculate record 6 by taking the weighted values of record 5 and so on.

PLATFORMS USED :

- Microsoft Excel
- Google Sheets
- Google Meet

CHALLENGES

As there were so many records, we had to wait for each person to complete their part so that we could do the calculations for assigned parts. The process of calculation was quite lengthy and time consuming.

RESULTS

Given Training Set :

For 12 Records

Training set	Record 1	Record 2	Record 3	Record 4	Record 5	Record 6	Record 7	Record 8	Record 9	Record 10	Record 11	Record 12
X0	1	1	1	1	1	1	1	1	1	1	1	1
X1	0.06	0.72	0.59	0.97	0.46	0.14	0.65	0.29	0.84	0.19	0.89	0.91
X2	0.3	0.63	0.08	0.18	0.08	0.82	0.53	0.02	0.22	0.09	0.83	0.4
X3	0.07	0.05	0.18	0.92	0.07	0.08	0.38	0.18	0.5	0.59	0.24	0.41
X4	0.84	0.51	0.81	0.54	0.45	0.16	0.44	0.76	0.4	0.88	0.39	0.75
Target variable	0.51	0.68	0.52	0.75	0.45	0.41	0.43	0.47	0.66	0.54	0.56	0.84

Calculated Weights : For 12 Records

Weights	Initial values													
W_1A	0.48	0.47999344	0.479980828	0.47992152	0.47993069	0.479865675	0.4798472224	0.4798372901	0.4798342648	0.4798312981	0.44479830289	0.4479852589	0.44479820	
W_1B	0.5	0.49999348	0.499979853	0.4999254	0.49993331	0.499871633	0.4998465422	0.49992421	0.4998465422	0.4998214514	0.4989892192	0.4998214514	0.499796	
W_2A	0.9	0.8999672	0.899956161	0.89994812	0.89994982	0.899938514	0.8998303437	0.8998295957	0.8998288527	0.8998298527	0.8999298527	0.8999398527	0.899949	
W_2B	0.24	0.23996739	0.239955472	0.23994809	0.23994956	0.239938829	0.2397918686	0.2397910756	0.2397900966	0.2397910966	0.2397911966	0.2397912966	0.239791	
W_3A	0.4	0.39999235	0.39999147	0.39997338	0.39998207	0.399852039	0.3998414947	0.3998407097	0.3998406185	0.3998405185	0.3998404185	0.3998304185	0.399829	
W_3B	0.26	0.25999239	0.259991446	0.25997483	0.25998234	0.259972949	0.2599586114	0.2599576244	0.2599567334	0.2599558424	0.2599549514	0.2599540604	0.259953	
W_4A	0.07	0.06990815	0.069899216	0.0698178	0.0698229	0.069759298	0.0697382093	0.0697373333	0.0697372352	0.0697311371	0.069730390	0.0697369409	0.069736	
W_4B	0.17	0.1699087	0.169899052	0.1698243	0.1698287	0.169768364	0.1697396888	0.1697388128	0.1697380488	0.1696777128	0.1696490376	0.1696481616	0.169647	
W_0A	0.28	0.27989065	0.279873138	0.27977262	0.27978207	0.279640732	0.279508128	0.279500188	0.279638384	0.279359644	0.279212784	0.2792191	0.27908	
W_AC	0.8	0.79996193	0.799953384	0.79991697	0.79992172	0.799873608	0.7998111379	0.7998101509	0.7998092749	0.7998082879	0.7998074119	0.7998064249	0.799805	
W_AD	0.86	0.85943916	0.859429772	0.8593895	0.85939469	0.859341214	0.8592712483	0.8592703573	0.8592702589	0.8592693679	0.8592692695	0.8592683785	0.859268	
W_0B	0.66	0.65989131	0.659872389	0.6597801	0.65978826	0.65965417	0.65949245	0.659474079	0.6594739919	0.659294859	0.6594971209	0.6592947119	0.65929	
W_BC	0.59	0.58995875	0.589950163	0.58990954	0.58991448	0.589914977	0.58986169	0.58977499	0.58976656	0.58967986	0.58967143	0.58958473	0.589	
W_BD	0.78	0.77995402	0.779944584	0.7798966	0.77990506	0.779845125	0.77978545	0.77977756	0.77976992	0.77976203	0.77975439	0.7797465	0.7797	
W_0C	0.75	0.74756291	0.747551915	0.74749854	0.74750456	0.747430875	0.74734818	0.74737164	0.74733272	0.74732618	0.74731726	0.74731072	0.7473	
W_CZ	0.52	0.51928199	0.519128841	0.51843889	0.51852532	0.517604156	0.51654161	0.51653178	0.51652246	0.51651263	0.51650331	0.51649348	0.5164	
W_0D	0.55	0.54993575	0.549923667	0.54986463	0.54987122	0.549789315	0.54969667	0.54959561	0.5495178	0.5494172	0.5493389	0.5492383	0.54	
W_DZ	0.57	0.56928472	0.569131633	0.56844297	0.56859426	0.567611123	0.56655347	0.5665436	0.5664647	0.56645483	0.56637593	0.56636606	0.5662	
W_0Z	0.03	0.02915091	0.028973186	0.02816241	0.02826226	0.027173067	0.02592248	0.02591459	0.0259067	0.02589881	0.02589092	0.02588303	0.0258	

For 12 to 24

0.4479820289	0.4479815289	0.4479810289	0.4479805289	0.4479800289	0.4479795289	0.4479789289	0.4479782289	0.4479774289	0.4479767289	0.4479761289	0.4479753289	0.4479746289
0.4997963606	0.4997941284	0.4997963606	0.4997712698	0.4998490376	0.4997712698	0.499746179	0.4998239468	0.499746179	0.4997210882	0.4997988856	0.4997210882	0.4996959974
0.8999498527	0.8999598527	0.8999698527	0.8999798527	0.8999898527	0.8999998527	0.8999999527	0.8999999627	0.8999999727	0.8998900098	0.899891098	0.899891928	0.8998952968
0.2397913966	0.2397914066	0.2397915066	0.2397916066	0.2397916166	0.2397917066	0.2397918066	0.2397928066	0.2397929066	0.2397930066	0.2397930166	0.2397940166	0.2397950166
0.3998294185	0.3998284185	0.3998274185	0.3998273185	0.399827413	0.399837413	0.399836413	0.399835413	0.399834413	0.399833413	0.399833313	0.399833213	0.399823213
0.2599531694	0.2599522784	0.2599513874	0.2599504964	0.2599496054	0.2599487144	0.2599478234	0.2599469324	0.2599460414	0.2599451504	0.2599442594	0.2599433684	0.2599424774
0.0697368428	0.0697367444	0.0697366466	0.0697365485	0.0697364504	0.0697363523	0.0697362542	0.0697361561	0.0697360580	0.0697359599	0.0697358618	0.0697357637	0.0697356656
0.1696473976	0.1695870616	0.1695583864	0.1695575104	0.1695567464	0.1696470164	0.1696477352	0.1696468592	0.1696460589	0.1694057592	0.169377084	0.169376208	0.169374544
0.279087296	0.279078556	0.278946752	0.278938012	0.278806208	0.278797468	0.278665664	0.278656924	0.27852512	0.27851638	0.278384576	0.278375836	0.278244032
0.799805489	0.7998045619	0.7998036599	0.7998026989	0.7998018229	0.7998008359	0.7997999599	0.7997989729	0.7997980699	0.7997971099	0.7997962339	0.7997952469	0.7997943709
0.8592682801	0.8592673891	0.8592673807	0.8592663997	0.8592663013	0.8592654103	0.8592653119	0.8592644209	0.8592643225	0.8592643415	0.8592633331	0.8592624421	0.8592623437
0.659293988	0.6594730338	0.659115639	0.659472249	0.6592946848	0.6591148551	0.6596512957	0.65893729	0.659471466	0.6594737306	0.6587574603	0.6600079066	0.6587592499
0.5895763	0.5894896	0.58948117	0.58939447	0.58938604	0.58929934	0.58929091	0.58920421	0.58919578	0.58910908	0.58910065	0.58901395	0.58900552
0.77973886	0.77973907	0.77972333	0.77971544	0.77970708	0.77969991	0.77969227	0.77968438	0.77967674	0.77966885	0.77966121	0.77965332	0.77964586
0.7473018	0.74729526	0.74728634	0.74727598	0.74727088	0.74726434	0.74725542	0.74724888	0.74723996	0.74723432	0.7472245	0.74722196	0.74720904
0.51648416	0.51647433	0.51646501	0.51645518	0.51644586	0.51643603	0.51642671	0.51641688	0.51640756	0.51639773	0.51638841	0.51637858	0.51636926
0.54916	0.5490594	0.5489811	0.5488805	0.5488022	0.5487016	0.5486233	0.5485527	0.5484444	0.5483488	0.5482655	0.5481649	0.5480866
0.56628716	0.56627729	0.56619839	0.56618852	0.56610962	0.56609975	0.56602085	0.56601098	0.56593208	0.56592222	0.56584331	0.56583344	0.56575454
0.02587514	0.02586725	0.02585936	0.02585147	0.02584358	0.02583569	0.02582878	0.02581991	0.02581202	0.02580413	0.02579624	0.02578835	0.02578046

For 25 to 37

0.4479737289	0.4479732289	0.4479725969	0.4479718649	0.4479712219	0.4479705789	0.4479697469	0.4479688469	0.4479680639	0.4479671909	0.4479664379	0.4479659489	0.4479647159
0.4997737652	0.4996959974	0.4996709066	0.4997486744	0.4996709066	0.4996458158	0.4997235836	0.4996458158	0.499620725	0.4996684928	0.499620725	0.4995956342	0.4996734032
0.899898298	0.899968298	0.899968298	0.899976298	0.899984298	0.899898498	0.8998985198	0.8998985998	0.899929298	0.899993798	0.899994078	0.899806342	0.899813432
0.2397951166	0.2397961166	0.2397962166	0.2397972166	0.2397973166	0.2398073166	0.2398083166	0.2398183166	0.2398193166	0.2398203166	0.2398213166	0.2398313166	0.239831216
0.399732213	0.399772213	0.399772213	0.3997902213	0.399692213	0.399952213	0.399522113	0.399582213	0.399572113	0.399562113	0.399552113	0.39852113	0.398542213
0.2599415864	0.2599406594	0.2599398044	0.2599389134	0.2599380224	0.2599371314	0.2599362404	0.2599353494	0.2599344584	0.2599335674	0.2599326764	0.2599317854	0.2599308944
0.0697356575	0.0697354694	0.0697353713	0.0697352732	0.0697351751	0.0697350706	0.0697349789	0.0697348808	0.0697347827	0.0697346846	0.0697345865	0.0697344884	0.0697343903
0.169315108	0.1692864328	0.169285568	0.1692847928	0.1692245468	0.1691957816	0.1691949056	0.1691941416	0.1691338056	0.1691051304	0.1691042544	0.1691034904	0.1690431544
0.27823592	0.278103488	0.278094748	0.277962944	0.277795404	0.2778216	0.27781366	0.277681856	0.277673116	0.277541316	0.277532572	0.277400768	0.277392028
0.7997933839	0.7997925079	0.7997915209	0.7997906449	0.7997896579	0.7997887819	0.7997877949	0.7997869189	0.7997859319	0.7997850559	0.7997840678	0.7997831929	0.7997822059
0.8592614527	0.8592613543	0.8592604633	0.8592603649	0.8592594739	0.8592593755	0.8592584845	0.8592583861	0.8592574951	0.8592573967	0.8592565057	0.8592564073	0.8592555163
0.6592916363	0.6600010712	0.6580434564	0.6605420826	0.6587619885	0.658575366	0.6612606175	0.6567952729	0.661073994	0.6594805244	0.6566086494	0.6637592455	0.6550151798
0.58981886	0.589891039	0.58882369	0.58881526	0.58872856	0.58872013	0.58863344	0.588625	0.58859383	0.58859287	0.58844317	0.58843474	0.58834084
0.77963779	0.77963015	0.77962226	0.77961462	0.77960673	0.77959909	0.77959112	0.77958356	0.77957567	0.77956803	0.77956014	0.77955525	0.77954461
0.7472025	0.74719358	0.74718704	0.74717812	0.74717158	0.74716266	0.74715612	0.74714606	0.74713174	0.74712532	0.74711628	0.747110974	
0.51639453	0.51635011	0.51634028	0.51633066	0.51632113	0.51631181	0.51630918	0.51629266	0.51628283	0.51627351	0.51626368	0.51624545	
0.547986	0.5479077	0.5478071	0.5477288	0.5476282	0.5475499	0.5474493	0.547371	0.5472704	0.5471921	0.5470915	0.5470132	0.5469126
0.56574467	0.56566577	0.5656559	0.565577	0.56556713	0.56548823	0.56547836	0.56539946	0.56538959	0.56531069	0.56530082	0.56522192	0.56521005
0.02577257	0.02576468	0.02575679	0.0257489	0.02574101	0.02573312	0.02572523	0.02571734	0.02570945	0.02570156	0.02569367	0.02568578	0.02567788

For 38 to 50

0.4479636929	0.4479630389	0.4479623909	0.4479609079	0.4479598739	0.4570017978	0.470017978	0.470012305	0.470002462	0.470005949	0.470002966	0.470000483	0.46999864
0.4995956342	0.4995705434	0.4996483112	0.4995705434	0.4995454526	0.4996232204	0.4995454526	0.4995203618	0.4995981296	0.4995203618	0.499495271	0.4995730388	0.499495271
0.899873432	0.89989999	0.89990999	0.89991099	0.89991199	0.89992199	0.89995852	0.89995752	0.89995552	0.89995252	0.89994852	0.89994252	0.89993851
0.239841216	0.239842216	0.239843216	0.24883882	0.24883982	0.24884982	0.24885082	0.24885182	0.23995883	0.23994884	0.23994554	0.23993994	0.23993883
0.398532113	0.398522113	0.398512113	0.398502113	0.398492113	0.398482113	0.398472113	0.39988204	0.39987204	0.39986233	0.39986133	0.39985304	0.39985204
0.2599300034	0.2599291124	0.2599282214	0.2599273304	0.2599264394	0.2599255484	0.25999295	0.25998295	0.25998125	0.25997995	0.25997698	0.25997376	0.25997295
0.0697342922	0.0697341941	0.0697340960	0.0697339979	0.0697338998	0.0697338017	0.0697337036	0.0699483	0.0699273	0.0698763	0.0698288	0.0697993	0.0697593
0.1690144792	0.1690136032	0.1690128392	0.1689525032	0.168923828	0.168922952	0.168922188	0.168861852	0.1688331768	0.1688323008	0.1688315368	0.1687712008	0.1687425256
0.277260224	0.277251484	0.27711968	0.27711094	0.276979136	0.276970396	0.276838592	0.276829852	0.276698048	0.276689308	0.276557504	0.276548764	0.27641696
0.7997813299	0.7997803429	0.7997794669	0.7997784799	0.7997776039	0.7997766169	0.7997757409	0.7997747539	0.7997738779	0.7997728909	0.7997720149	0.7997710279	0.7997701519
0.8592554179	0.8592545269	0.8592544285	0.8592535375	0.8592534391	0.8592525481	0.8592524497	0.8592515587	0.8592514603	0.8592505693	0.8592504709	0.8592495799	0.8592494815
0.6608873705	0.6621657759	0.6521433048	0.6680379666	0.6534217102	0.6580154955	0.669316372	0.6433992391	0.6739101573	0.6547001156	0.6479930244	0.6852110338	0.6287829827
0.58833961	0.58825291	0.58824448	0.58815778	0.58814935	0.58806265	0.58805422	0.58796752	0.58795909	0.58787239	0.58786396	0.58777726	0.58776883
0.77953697	0.77952908	0.77952144	0.77951355	0.77950591	0.77949802	0.77949038	0.77948249	0.77947485	0.77946696	0.77945932	0.77945143	0.77944379
0.74710082	0.74709428	0.74708536	0.74707882	0.7470699	0.74706336	0.74705444	0.7470479	0.74703898	0.74703244	0.74702352	0.74701698	0.74700806
0.51623521	0.51622538	0.51621606	0.51620623	0.51619691	0.51618708	0.51617776	0.51616793	0.51615861	0.51614878	0.51613946	0.51612963	0.51612031
0.5468343	0.5467337	0.5466554	0.5465548	0.5464765	0.5463759	0.5462976	0.546197	0.5461187	0.5460181	0.5459398	0.5458392	0.5457609
0.56513315	0.56512328	0.56504438	0.56503451	0.56495561	0.56494574	0.56486684	0.56485697	0.56477807	0.5647682	0.5646893	0.56467943	0.56460053
0.02567	0.02566211	0.02565422	0.02564633	0.02563844	0.02563055	0.02562266	0.02561477	0.02560688	0.02559899	0.0255911	0.02558321	0.02557532

Credits :

1. Abhishek Katti
2. Bhavana Jayaram
3. Sri Ganesh Reddy
4. Sai Yaswanth